Q10 AI DEMONSTRATION

TelcoX - Real-Time versus Batch Processing

A presentation-ready demonstration using the supplied Excel workbook

What this demo proves

The AI can inspect TelcoX data, recommend which signals need rapid processing, separate those from scheduled batch workloads, and explain what the workbook can and cannot prove. The team then verifies the AI output against the source sheets and calculations.

Dataset used

Sheet	Records	Relevant fields
Customers	200	customer_id, monthly_spend, tenure_months, plan_type, nps_score, churn_6mo_flag
Calls	5,000	customer_id, call_timestamp, call_reason_tag, resolved_flag, hold_time_sec, transfer_count
NetworkEvents	1,000	cell_id, event_timestamp, event_type, loss_percentage, avg_latency_ms
Store_Visits	300	customer_id, visit_timestamp, visit_duration_min, issue_type

AI-generated recommendation

Use a HYBRID processing model: process network degradation and active service failures quickly; process customer profiles, historical churn labels, monthly trends and model training in scheduled batches.

Important boundary: the workbook is a historical dataset with event-occurrence timestamps. It does not contain streaming arrival times or ingestion timestamps. It can support the processing design, but it cannot prove current live-feed latency or demonstrate a production alert in real time.

1. Copy-paste prompt for the live AI demonstration

Upload the original workbook Case Study 1- DIAI framework - TelcoX Case Study Data.xlsx to ChatGPT or another AI assistant. Then paste the prompt below.

You are a data analytics assistant supporting TelcoX. Use ONLY the uploaded Excel workbook and the case facts provided. Do not invent fields, customer records, or business results.

1. Inspect the Customers, Calls, NetworkEvents, and Store_Visits sheets. State the row counts and the exact fields used.
2. Recommend which data signals should be processed in real time and which should be processed in batch. Explain why delayed processing reduces business value for each real-time signal.
3. Calculate and show: churn rate from churn_6mo_flag; resolved and unresolved call counts from resolved_flag; network event counts by event_type; counts where loss_percentage >= 10 and avg_latency_ms >= 200. Clearly label 10% and 200 ms as example screening thresholds, not company-approved limits.
4. Check ID uniqueness and missing IDs. Inspect timestamp formats and date ranges.
5. Identify missing fields or relationships that prevent customer-level attribution or true real-time monitoring.
6. Propose a simple hybrid architecture and practical pilot refresh/alert targets. Mark targets as recommendations, not observed facts.
7. End with a concise recommendation. Separate workbook facts, calculations, assumptions, and proposed actions. Cite the sheet and column names for every numerical claim.

2. Demonstration output - answer produced from the workbook

Recommendation: TelcoX should use a hybrid model. Network events and active customer-service failures need event-driven processing because the opportunity to prevent service disruption or customer frustration declines as the alert becomes older. Customer profiles, tenure, monthly spend, six-month churn labels, historical interaction patterns and trend reporting are suitable for batch processing.

Workbook evidence: The Customers sheet contains 200 customers, of whom 62 have churn_6mo_flag = 1, giving a 31.0% six-month churn rate. The Calls sheet contains 5,000 calls: 3,986 resolved (79.72%) and 1,014 unresolved (20.28%). NetworkEvents contains 1,000 records: 340 packet_loss, 336 latency_spike and 324 throughput_drop events.

Operational screening: In the supplied network rows, 463 events have loss_percentage >= 10 and 398 have avg_latency_ms >= 200. These are useful examples for an alert-rule demonstration, but the workbook does not say that 10% or 200 ms are approved TelcoX thresholds. Engineering must validate thresholds and persistence windows before production use.

Data limitations: Calls and Store_Visits contain customer_id, so they can link to Customers. NetworkEvents contains cell_id but no customer_id or serving-cell/session mapping. Therefore, cell-level incidents cannot safely be attributed to named customers from this workbook alone. Store_Visits has visit_duration_min but no live queue or wait_time_minutes field. All three timestamp columns mix text dates and Excel serial values. No ingestion timestamp is available, so end-to-end processing delay cannot be measured.

Practical design: Send network and active call events through a fast validation and alert path. Route validated records into a historical warehouse as well. Run nightly batch integration for customer, call, network and store history; refresh churn features and dashboards daily; train/review models on a scheduled cycle. A suggested pilot target is under 1 minute for severe network-event alerting and under 5 minutes for failed support-interaction alerts, subject to operational testing.

3. Verify the AI output against the actual workbook

Do not accept the AI answer just because it sounds reasonable. Open the named sheet, inspect the field, and reproduce the calculation. The figures below were independently recomputed from the supplied workbook.

AI claim	Workbook check	Verified result
Customer churn is 31.0%	Customers!H2:H201; count churn_6mo_flag = 1 and divide by 200 valid rows.	62 / 200 = 31.0%
Calls: 79.72% resolved, 20.28% unresolved	Calls!G2:G5001; count 1 and 0 values.	3,986 resolved; 1,014 unresolved.
Network-event mix	NetworkEvents!D2:D1001; count each event_type.	340 packet_loss; 336 latency_spike; 324 throughput_drop.
Loss screening count	NetworkEvents!E2:E1001; count values >= 10.	463 events (46.3%). Threshold is illustrative.
Latency screening count	NetworkEvents!F2:F1001; count values >= 200.	398 events (39.8%). Threshold is illustrative.
Unique IDs	Check customer_id, call_id, event_id and visit_id for blanks and duplicates.	200/200 customer IDs unique; all 5,000 call IDs, 1,000 event IDs and 300 visit IDs unique; no missing IDs found.

Timestamp and relationship checks

Sheet	Mixed timestamp storage	Parsed date range
Calls	2,879 text values; 2,121 Excel serial values.	2025-01-06 to 2025-12-06
NetworkEvents	581 text; 418 Excel serial; 1 integer serial.	2025-01-06 to 2025-12-06
Store_Visits	177 text values; 123 Excel serial values.	2025-01-06 to 2025-12-06

Interpretation: all timestamp values in these sheets could be parsed, but they are stored in mixed formats. Standardise them to one timestamp format and timezone before time-window processing. The workbook covers historical 2025 activity; it is not a current live feed. Because event-ingestion time is missing, do not report a measured processing latency or claim that a production real-time alert has been tested.

4. Recommended model and architecture

FAST / EVENT-DRIVEN PATH	SCHEDULED / BATCH PATH
NetworkEvents: event_type, cell_id, loss_percentage, avg_latency_ms, event_timestamp Active Calls: customer_id, call_reason_tag, resolved_flag, hold_time_sec, transfer_count Action: validate, enrich, detect threshold/persistence breach, notify NOC or supervisor, and log the incident.	Customer master: plan_type, tenure_months, monthly_spend, nps_score, churn_6mo_flag History: completed calls, completed store visits, network-event history Action: standardise timestamps, deduplicate, join records, calculate daily/weekly features, refresh dashboards and train/review models.

Important integration controls

Customer joins: use customer_id to connect Customers with Calls and Store_Visits. Network joins: use cell_id for cell-level analysis. To connect network incidents to customers, obtain a time-aware serving-cell/session mapping; do not join cell_id directly to customer_id. Time control: preserve event time, add ingestion time, standardise timezone, and monitor late-arriving events.

Pilot targets (proposed, not measured)

Signal	Suggested operating target	Why
Severe network event	Alert within 1 minute after validation and persistence check.	Enables faster network investigation and customer notification.
Failed / unresolved support interaction	Supervisor alert within 5 minutes.	Allows service recovery before the customer repeats contact or escalates.
Customer/churn analytics	Nightly refresh; monthly trend review.	Historical data changes more slowly and is suitable for scheduled aggregation.

5. Three-minute presentation script

0:00-0:30 | Introduce: “We asked AI to examine the four TelcoX sheets and decide which information loses value when processing is delayed. We asked it to use the workbook only and not invent information.”

0:30-1:15 | Run the prompt: Upload the workbook, paste the prompt from page 2, and show the AI response. Point out the recommendation for a hybrid approach: event-driven processing for network/service incidents and batch processing for historical customer analytics.

1:15-2:15 | Verify: Open Customers and show 62 churn flags among 200 customers. Open Calls and show 3,986 resolved and 1,014 unresolved calls. Open NetworkEvents and count event types; then filter loss_percentage >= 10 and avg_latency_ms >= 200. Compare the workbook counts with the AI answer.

2:15-2:45 | Explain limitations: “AI also identified that network rows have cell_id but not customer_id, store visits do not have a live wait-time field, and timestamps mix text with Excel serials. We checked these limitations directly in the columns.”

2:45-3:00 | Conclude: “Our recommendation is not to make every dataset real time. We process urgent operational events quickly and use batch processing for slower-changing history. We would add ingestion timestamps, standardise time, and build the missing cell-to-customer mapping before production.”

Do not claim: that the workbook is a live stream, that the suggested alert thresholds are approved TelcoX policies, that customer identities can be inferred from cell_id, or that processing latency has been measured. Those claims are not supported by the supplied data.